

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true of Church's thesis?
 - (a) It was first proven by Alan Turing.
 - (b) It has not yet been proven, but finding a proof is a subject of active research.
 - (c) It can never be proven.
 - (d) It is now in doubt because of the advent of parallel computers.
3. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x^{**}2]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100
4. What is the output of `"abc"[::-1]` in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
5. The boolean expression $a[i] == \max \parallel \neg(\max \neq a[i])$ can be simplified to
 - (a) $a[i] == \max$
 - (b) $a[i] \neq \max$
 - (c) $a[i] < \max \parallel a[i] > \max$
 - (d) FALSE

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][1]` evaluates to `'a'`.
7. True or False: Authentication is compromised when an unauthorized user accesses a system with stolen login credentials.
8. True or False: Reference counting requires less space overhead compared to other garbage collection methods.
9. True or False: The function `sorted('ab')` returns a new sorted list of the characters in the string `'ab'`.
10. True or False: Static allocation is sufficient for all types of functions, including those that utilize recursion.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether a sequence of numbers has an increasing trend or not.
12. Write a function to check if the given tuple has any none value or not.

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